

Zhuohao Chen

Purdue University
Department of Computer Graphics Technology
M.S.

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Education

M.S. Computer Graphics Technology, Purdue University, West Lafayette, IN Expected May 2026
GPA: 4.0/4.0

B.Eng. Computer Science and Technology, Changsha University of Science and Technology, Changsha, China June 2024
GPA: 86.68/100

Fields of Interests

Human-Computer Interaction (HCI) & Novel Interaction Techniques, Mobile, Wearable, and Ubiquitous Computing, Context-Aware AR & Adaptive Interfaces, Behavioral Sensing & Input Methods in XR, Accessibility & Perceptual Augmentation.

Professional Memberships

Association for Computing Machinery (ACM), 2024 – Present

ACM SIGCHI (Special Interest Group on Computer-Human Interaction), 2024 – Present

IEEE (Institute of Electrical and Electronics Engineers), 2024 – Present

Publications

Chen, Z.*, Wu, Z.*, & Mousas, C. Exploring Perception and Avoidance Behavior toward Real and Virtual Humans in Outdoor Mixed Reality. In Proceedings of the 4th Workshop on Locomotion and Wayfinding in XR (Low-XR) at IEEE VR 2026. (Accepted; *Co-first authors)

Wu, Z., **Chen, Z.**, Zhu, D., Mousas, C., & Kao, D. (2025). A Systematic Review of Generative AI on Game Character Creation: Applications, Challenges, and Future Trends. IEEE Transactions on Games, Early Access. <https://ieeexplore.ieee.org/document/10979443>

Yan, J., Sun, J., Tang, T., & **Chen, Z.** (2023, July). Visual Design for Predictive Display in Spatial Time-Delay Environments Considering Workload. In International Conference on Human-Computer Interaction (pp. 326-340). Cham: Springer Nature Switzerland.

Chen, Z. (2023). Research and Design of Chinese Cookie Model with Chun Character based on OpenSCAD. Accepted at CONF-CDS 2023.

Research Projects

Graduate Researcher, Department of Computer Graphics Technology,

Purdue University

2025 – Present

- Investigating perception, avoidance behavior, and psychological equivalence of real and virtual obstacles in outdoor MR environments.
- Designed and conducted controlled experiments using Meta Quest 3, including measures of spatial presence, reaction time, movement trajectory, and cognitive load.
- Preparing manuscript *Comparative Analysis of Users' Perception, Avoidance Behavior, and Psychological Responses to Virtual and Real Obstacles in Outdoor MR Environments* (Submitted to IEEE VR 2026 / TVCG; Co-first author).

Co-Author, Graduate Research Project,

Purdue University

2024

- Conducted a PRISMA-based systematic review of 72 studies on generative AI in game character design.
- Developed statistical tables and visual classification frameworks; formatted the manuscript in LaTeX.
- Published as A Systematic Review of Generative AI on Game Character Creation: Applications, Challenges, and Future Trends, IEEE Transactions on Games (2024, Early Access).

Undergraduate Research Assistant, School of Computer Science and Technology,

Changsha University of Science and Technology (CSUST)

2022 – 2023

- Explored the effects of predictive display technologies (VR-PGI and I-PGI) on operator workload in remote human–computer interaction under varying time delays.
- Assisted in experimental design, participant recruitment (N=20), and data collection using a virtual simulation platform.
- Published as Visual Design for Predictive Display in Spatial Time-delay Environments Considering Workload, HCI International 2023 (Accepted).

Undergraduate Researcher, Changsha University of Science and Technology (CSUST)

Changsha University of Science and Technology (CSUST)

2022 – 2023

- Designed and implemented a 3D modeling approach using OpenSCAD to create a generative “Chun” character cookie model.
- Focused on parametric modeling and computational design methods for cultural and educational applications.
- Resulted in publication: Research and Design of Chinese Cookie Model with Chun Character based on OpenSCAD, CONF-CDS 2023 (Accepted).

Computer and Language Skills

C, C++, Java, Python, C#, Unity, Unreal Engine, OpenGL (GLSL), Blender, LaTeX, Git, Mandarin (Native), English (Fluent)

Honors and Fellowship

- Second Prize, China National Undergraduate Ruikang Robot Developer Competition, 2023
- Third Prize (Hunan Region), 24th China Robotics and AI Competition, 2022
- Second Prize, China Undergraduate Computer Design Competition, 2022
- Third Prize (Education Platform Track), China Undergraduate Computer Design Competition, 2022
- Third Prize, China Undergraduate Innovation and Entrepreneurship Competition, 2022
- First Prize, 3rd “Yuanguang Cup” Game Creation Competition (CSUST), 2022

Leadership Involvement

Leader, Game Development Club, CSUST

July 2022 – September 2023

- Organized and led weekly workshops on Unity and C# for undergraduate students.
- Coordinated peer-learning sessions and collaborative projects to strengthen programming skills.
- Mentored junior members in game design, providing feedback on coding practices and mechanics.

Founder & Initiator, “Yuanguang Cup” Game Creation Competition, CSUST

November 2022 – December 2022

- Established the university’s first student-led game design competition.
- Coordinated with faculty, student organizations, and sponsors to organize the event.
- Recruited over 50 participants, fostering interdisciplinary collaboration in game development.

Relevant Experience

Undergraduate Research Assistant

School of Computer Science and Technology

CSUST, Changsha, China

May 2023 – June 2023

- Developed an interactive VR museum system using Unity and HTC Vive.
- Implemented user interaction logic and immersive environment features to support usability studies.
- The system was later adopted in graduate-level user research and contributed to a peer-reviewed publication.